Response Letter with Annotations

Li, K., Mai, F., Shen, R., & Yan, X. (2021). Measuring corporate culture using machine learning. *The Review of Financial Studies*, *34*(7), 3265-3315.

Annotations

1. Purposes

a. Motivation (one-sentence summary)

To provide a semisupervised machine learning approach using the word embedding model in quantifying text on the finance and accounting literatures to measure corporate culture.

b. Specific Rationale (several bullet points)

-There are limitations of supervised learning or data driven analysis regarding text analysis, the approach of this paper complements those limitations, as their semisupervised method can be used when it is not possible to use either supervised learning or data driven analysis

-They used earnings calls because they represent corporate values well

-The dictionary they built from data makes it hard to artificially emphasize one value over others.

c. Research Questions and/or Hypotheses

Although they did not explicitly mention either RQs or Hypotheses,

1. They aimed to test whether their methods using a new semisupervised machine learning approach, word2vec, can be a proper approach to measure corporate culture.
2. They aimed to explore the impact of (strong) corporate cultural values on business outcomes.

2. Method

a. Research Design

Computational text analysis research

1. Train the word embedding model and then obtain corporate cultural values for 7,501 unique firms (62,664 firm-year observations)

- Generating the culture dictionary using
- Scoring corporate culture

2. Validation

- Validate these obtained corporate culture measures using well-established markers for each of corporate innovation, integrity, product quality, respect, and teamwork.
- Validate obtained measure by comparing it with alternative measures.

4. Examine the influence of corporate culture on business outcomes.

b. Sample

- They used 209,480 QA sections of earnings call transcripts from Thomson Reuters' StreetEvents (SE) database for the period January 1, 2001, to May 25, 2018.
- The original data contained the ticker symbol header, the company name, the title of the event, and the date of the call. After matching, the earnings call data set consists of 209,480 QA sections that can be mapped to 64,511 firm- year observations.

c. Measures

Corporate cultural value:

They made a culture dictionary with value words to measure corporate cultural value using value word/seed words and scored each value.

"Value word": Word that expresses a core corporate value, which are innovation, integrity, quality, respect, and teamwork

Measure the five cultural values at the firm–fiscal year level using weighted word frequencies divided by the total number of words in the document.

Markers for 5 values

Innovation: ln(Patent), R&D spending, and innovation strength

Integrity: Restatement, Backdating

Quality: Product quality, product safety, and top brand.

Respect: Diversity and "best employer" status.

Teamwork: the level of employee involvement and the number of joint ventures (JVs) and strategic alliances (SAs).

Business outcome:

They used operational efficiency, risk-taking, the incentive to manage earnings, executive compensation design, and firm value for business outcome variables.

d. Analysis

Data, preprocessing and parsing, and learning phrases

-Stanford CoreNLP package to preprocess and parse the text

-Word Embedding, word2vec, and Model Training

Word2vec: they used the gensim library in Python to train the word2vec model.

The validation

They validate each value with markers of each one.
- When the dependent variables are ln(Patent), R&D spending, diversity, and the number of JVs/SAs: **Ordinary least squares (OLS) regressions.**
- For all other validating variables: **Probit regressions.**

Scoring corporate culture value

They trained the word2vec model to develop an expanded, context-specific dictionary. They manually inspected all the words in the auto-generated dictionary and excluded words that do not fit.

Then, they measured each of the five cultural values at the firm-fiscal year level. The firm-year measures of cultural values are based on 3-year moving averages of annual values. They weighted the count of the number of words associated with each value divided by the total number of words in the document (The weight is tf.idf).

Comparing with other alternative methods of measuring corporate culture

Using alternative measures and run the same validation test using OLS regressions and Probit regressions.

Examine impact on business outcomes
OLS regression

3. **Results and Conclusions**
    a. Answers to RQs and Hypotheses

1. **Their methods using a new semisupervised machine learning approach, word2vec, can be a proper approach to measure corporate culture: Supported**

-Validation markers: measure of corporate culture is correlated with shared values and practices by employees at large.

The cultural value of **innovation** is positively and significantly associated with all three measures of corporate innovation activities

The cultural value of **integrity** is negatively and significantly associated with one form of accounting malfeasance: restatement and backdating executives' option grants.

The cultural value of **quality** is positively and significantly associated with two out of three measures of product quality, namely, product safety and top brand status

The cultural value of **respect** is positively and significantly associated with the diversity score reported by KLD and the best employer ranking by Fortune.

The cultural value of **teamwork** is positively and significantly associated with both employee involvement from KLD and the number of JVs/SAs.

-Validate obtained measure by comparing it with alternative measures.

Their measure outperforms all these three alternative measures using MD&A section of 10-Ks, a simple alternative using the list of seed words full earning call .

2. **The impact of (strong) corporate cultural values on business outcomes:**

There was a strong association between corporate culture and business outcomes.

b. Include Key Figures/Tables
X
c. Authors' Conclusions

-This paper demonstrates that machine learning including word embedding, a natural language model based on artificial neural networks, can learn the context-specific meanings of words and phrases and can be useful for measuring corporate culture

-It shows that corporate culture plays a crucial role in business.

4. Discussion

a. Implications

-As far as they know it is among the first in the finance and accounting literatures, applying neural network language models to the analysis of qualitative corporate disclosures.

-By applying word2vec to earnings calls to score corporate cultural values, they examined the implications of having a strong corporate culture based on large sample evidence.

-By introducing the word embedding model as a new approach to quantifying the meaning of expressions and a new semisupervised machine learning approach for textual analysis, it contributes to literature on corporate culture spanning management, accounting, and finance, providing methods to measure corporate culture that can be easily applied to a large sample of firms over time.

b. Limitations

-The word embedding model has one limitation: multiple senses (meanings) of a word are combined into a single vector.
-Misclassification is inevitable in this method.

c. Future Directions

The paper suggests that machine learning holds promise for more applications in social sciences.

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Response Letter (~600 words)

- Your assessment of the study

I think this study is not only methodologically rigorous but also original in many ways. For example, when measuring corporate culture values, it is easy to use official documents as data. However, they explain and concern that "official documents" can include values that are artificially created or self-promoted. To identify companies' "real" values, this study uses the Q&A section of earnings calls. I found this approach very original and convincing.

I also felt that the study is methodologically rigorous because it conducts multiple validation tests. All methods, including computational methods, have limitations. In class, Professor Zhou taught us that we can strengthen findings by using different alternative methods and showing that the findings are consistent. This study did exactly like that. In addition to obtain values using Word2Vec, it also examines how strongly those values are related to real markers of each value, and then compares the results with values obtained from other alternative methods and datasets. I found this validation process very impressive, and I think it clearly strengthens the study.

Although the last part is less directly related to our computational methods class and I did not discuss it in detail above, the study also analyzes how corporate culture values identified through Word2Vec are related to real business outcomes, using many different variables. In my opinion, this part alone is strong enough to be a separate paper. Because the study shows how corporate culture values may influence actual business outcomes, I was able to clearly understand why we use text analysis, how computational methods like Word2Vec can be applied, and why they are important.

- Weaknesses

I did not find any critical weaknesses, but I do have some questions. All analyses in this paper are based on seed words based on the five most frequently mentioned values by S&P 500 firms on their corporate websites that previous study identified (Guiso, Sapienza, and Zingales, 2015). I am curious whether there might be other words that could also represent corporate culture values beyond these five, or better represent than these five.

I am also wondering whether this methodology is still widely being used, given the more advanced LLM models nowadays. One limitation of this approach is that word misclassification and multiple meanings of a word are combined into a single vector, and I am curious whether we use this method nowadays despite these limitations.

Beyond methodological concerns, I think that one weakness of the paper is its readability. The paper contains a very large amount of information, but it does not clearly state the research questions explicitly. In addition, the validation section includes many different types of analyses, which makes the paper harder to follow. As a reader, I had to read the paper multiple times to fully understand it.

Also, although the study makes a strong effort to capture corporate culture values like using QnA section, it might have been more rigorous if they had obtained other measures based on real human evaluations rather than values obtained through computational methods and had compared them. They mentioned that they considered using employee reviews, such as those from Glassdoor.com, but were not able to obtain the data. I think the study would have been even stronger if they could have obtained such data and further analyzed.